

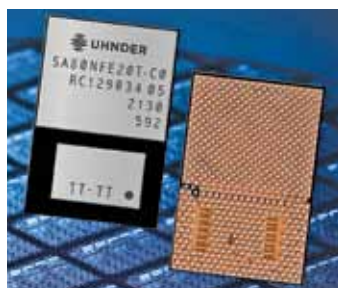
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Digital imaging radar

The 4D digital imaging radar-on-chip aims to enable next-generation advanced driver assistance systems, autonomous vehicles, and automated



mobility applications to seamlessly operate on roadways. Based on digital code modulation, the digital radar solution comes with better accuracy and power to sense moving or standing objects, large or small, at both short and long distances in all weather and lighting conditions, all while mitigating mutual interference between other radars. The digital radar can clearly identify a pedestrian crossing the road, a child entering the street from beside a parked car, or a bicyclist next to a guard rail, at levels of accuracy that legacy analogue radar systems cannot achieve.

Uhnder
www.uhnder.com

Wafer-on-wafer processor

The world's first 3D wafer-on-wafer (WoW) processor—the Bow IPU—delivers up to 40% higher performance and 16% better power efficiency for real world AI applications than its predecessors, all for the same price and requiring no changes to existing software. WoW bonds two wafers together to generate a new 3D die. The first wafer is for AI process-

ing and is architecturally compatible with the GC200 IPU processor. It has 1472 independent IPU-core tiles



and is capable of running more than 8800 threads. The second wafer is a power delivery die and features deep trench capacitors that enable a large performance increase thanks to being located right next to the processing cores and memory.

Graphcore
www.graphcore.ai

Molecular electronics chip

The first molecular electronics chip has been developed, realising a 50-year-old



goal of integrating single molecules into circuits to achieve the ultimate scaling limits of Moore's Law. The chip uses single molecules as universal sensor elements in a circuit to create a programmable biosensor with real-time, single-molecule sensitivity, and unlimited scalability in sensor pixel density. This innovation will power advances in diverse fields that are

fundamentally based on observing molecular interactions, including drug discovery, diagnostics, DNA sequencing, and proteomics.

Roswell Biotechnologies
www.roswellbiotech.com

Wireless SoC for satellite IoT

The ORC3990 is the first wireless system-on-chip (SoC) solution for the satellite Internet of Things (IoT) that meets the demands of satellite IoT connectivity. The SoC integrates all



required system functions into a single, cost-effective device in a very small 68-QFN package. An essential building block of the SoC design includes a flexible, third-generation Live Wireless RF, and digital radio subsystem. Additional function blocks integrated within the ORC3990 SoC include a low noise amplifier, a digital power amplifier, power management unit subsystem, including all analogue blocks, dual Arm Cortex-M0+ CPUs for separate network and application processing, all necessary memory (volatile and non-volatile) for the on-chip CPUs, required security functions, and key analogue and digital peripherals.

Orca Systems
www.orcasystems.com

Indoor light harvesting sensors

The ERS Eco and ERS Eco CO2 comprise a battery-free sensor line that uses Epishine's indoor light energy